

Marvell Semiconductor :

C Programming :

1. What is static variable?
2. What is global variable?
3. Explain flow from .c file to executable file?
4. What is volatile? Example of volatile?
5. What is extern? Extern declared in one file. Modified in another file. Will it have any effect in the file in which it is declared?
6. What is static function?
7. What is difference between definition and declaration of a function?
8. What is constant pointer, pointer to a constant?
9. What are bitwise and logical operators? Notations of them.
10. Write a code to set nth bit of a given number.
11. When segmentation fault occurs?
12. When memory leakage takes place? How to check memory leakage?
13. Snippet :

```
int arr[10];
int b=5;
arr = &b;
printf("%d",arr[0]);
```
14. Given :

```
int b = 32767;
```

➔ Store 40000 in b without changing the type size.
15. How to debug code using GDB?
16. Difference between structure and union?

Operating Systems :

1. What is virtual memory?
2. What is process?
3. What is thread? Difference between process and thread?
4. How is thread created?
5. If five threads are created in one process, what will happen if one of the thread gets terminated?
6. What are signals? Write code to implement signal handler?
7. Which signal is called by Ctrl+Z?
8. What are different IPC mechanisms?
9. What are sockets?
10. Write a code to implement socket?
11. What is semaphore, mutex, spinlock? Difference between them. Scenario when to use semaphore, mutex, spinlock. What are different types of semaphore?
12. Give example of binary semaphore and counter semaphore?
13. Explain process life cycle?
14. What is segmentation?
15. What is fragmentation?
16. Explain paging?

Networks :

1. What are different layers of OSI model?
2. What are switches and routers? Difference between them. In which layer they are used?
3. What is ping command? Purpose of ping command. Which protocol is used by ping command?
4. Explain TCP and UDP.

Device Drivers :

1. What is device driver?
2. What are character devices?
3. How do you load device driver (kernel module)?
4. What happens when you do insmod?
5. What is probe?
6. What are entry point function in device driver code?
7. What are fops?
8. How do you register, add, initialize a device?
9. What copy to user and copy from user do?
10. Tasklet
11. Work queue
12. Top half and bottom half
13. ioctl

Data Structure :

1. Where linked lists are used?
2. Find the loop in a singly linear linked list.

